

# SARS-CoV-2 methyltransferase nsp10-16 in complex with natural and drug-like purine analogs for guiding structure-based drug discovery

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## Abstract

Non-structural protein 10 (nsp10) and non-structural protein 16 (nsp16) are part of the RNA synthesis complex, which is crucial for the replication of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Nsp16 exhibits 2'-O-methyltransferase activity during viral messenger RNA capping and is active in a heterodimeric complex with enzymatically inactive nsp10. It has been shown that inactivation of the nsp10-16 protein complex interferes severely with viral replication, making it a highly promising drug target. As information on ligands binding to the nsp10-16 complex (nsp10-16) is still scarce, we screened the active site for potential binding of drug-like and fragment-like compounds using X-ray crystallography. The screened set of 234 compounds consists of derivatives of the natural substrate S-adenosyl methionine (SAM) and adenine derivatives, of which some have been described previously as methyltransferase inhibitors and nsp16 binders. A docking study guided the selection of many of these compounds. Here we report structures of binders to the SAM site of nsp10-16 and for two of them, toyocamycin and sangivamycin, we present additional crystal structures in the presence of a second substrate, Cap0-analog/Cap0-RNA. The identified hits were tested for binding to nsp10-16 in solution and antiviral activity in cell culture. Our data provide important structural information on various molecules that bind to the SAM substrate site which can be used as novel starting points for selective methyltransferase inhibitor designs.

### eLife assessment

This **important** study describes the crystallographic screening of a number of small molecules against a viral enzyme critical for the 5' capping of SARS-CoV-2 RNA and viral replication. While the high-quality crystal structures and complementary biophysical assays in this study provide **solid** evidence to support the major claims regarding how these small molecule compounds bind to the viral enzyme, the mismatch between the antiviral activity and binding to the viral enzyme of several small molecule compounds could have been more thoroughly investigated or discussed. This paper would be of interest to the fields of coronavirus biology, structural biology, and drug discovery.

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## Introduction

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) can cause a mild to severe and life-threatening illness, termed coronavirus disease 2019 (COVID-19). It is similar to the disease caused by SARS-CoV and middle east respiratory syndrome coronavirus (MERS-CoV). Due to the global spread of SARS-CoV-2 and its impact on public health a call to action for a coordinated response worldwide was required. The World Health Organization (WHO) declared a pandemic in March 2020 (Zhou et al. 2020 [↗](#)). Over 7 million cumulative deaths have been confirmed up to the end of 2023 (WHO 2023). Effective vaccines that reduced the number of critically ill patients needing hospitalization (Sandmann et al. 2021 [↗](#)) were developed fast, however, immune escape variants have emerged (Dong et al. 2020 [↗](#); Parums 2023 [↗](#)). Continuous efforts are in place to adapt these vaccines to address emerging variants (Grant et al. 2023 [↗](#)). Approved drugs for COVID-19 treatment are available as well. However, they are mainly in use to treat patients at high risk of adverse outcomes due to the high costs, possible side effects and the potential emergence of drug-resistant strains (Pachetti et al. 2020 [↗](#); Food and Administration (FDA) 2021 [↗](#); 2021; Parums 2022 [↗](#); Bernal et al. 2022 [↗](#); Chatterjee et al. 2023 [↗](#); Parums 2023 [↗](#)). Drugs directed toward highly conserved viral proteins are more likely to act on variants of concern unresponsive to vaccine-induced protection and are less likely to provoke the genesis of treatment escape variants (Pachetti et al. 2020 [↗](#); Chatterjee et al. 2023 [↗](#); Parums 2023 [↗](#)).

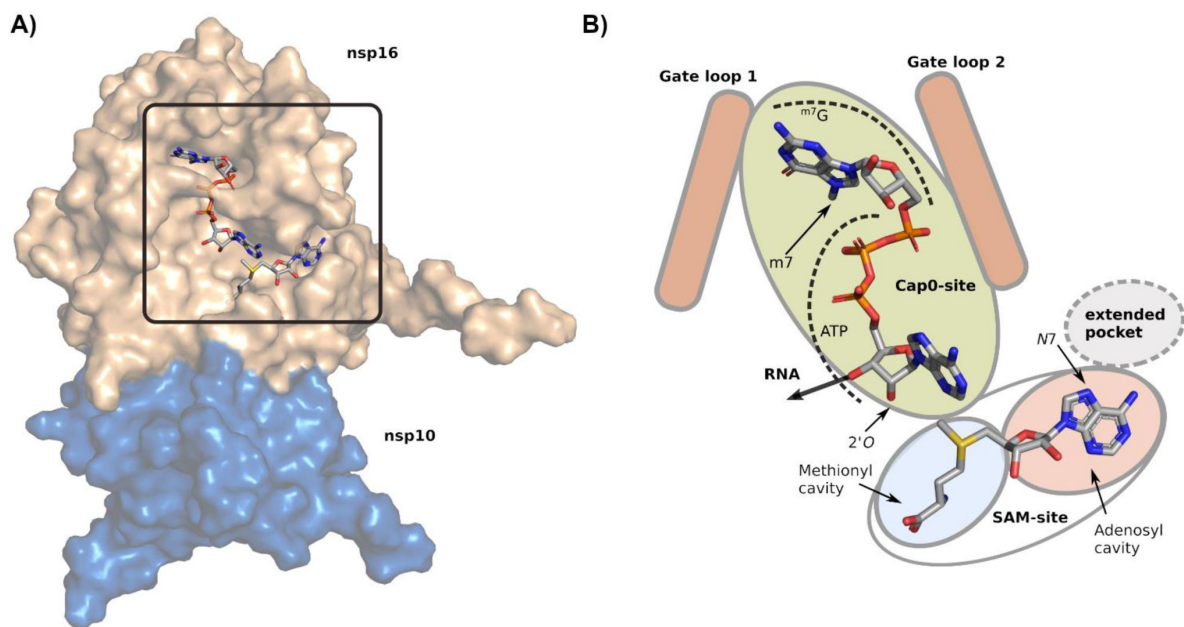
SARS-CoV-2 is an enveloped  $\beta$ -coronavirus with a large (30 kb), complex, positive-sense single-stranded RNA genome. The genomic RNA contains a 5'-cap and a 3'-poly-A tail similar to host mRNAs, thereby preventing immune system activation (Menachery, Debbink, and Baric 2014 [↗](#); Daffis et al. 2010 [↗](#); Chang and Chen 2021 [↗](#); Bradrick 2017 [↗](#)), enabling recognition by the host translation machinery, and increasing stability (Zhu et al. 2020 [↗](#); Furuichi and Shatkin 2000 [↗](#); Schwer, Mao, and Shuman 1998 [↗](#); Cougot et al. 2004 [↗](#); Lewis and Izauride 1997 [↗](#)). The 5' capping process involves several viral non-structural proteins (nsp) such as nsp13, RNA triphosphatase (TPase) helicase (Bouvet et al. 2010 [↗](#)), the RNA guanylyltransferase (GTase) nsp12 (Walker et al. 2021 [↗](#); Yan et al. 2021 [↗](#)), the N7-methyltransferase (N7-MTase) which is a complex of nsp10 and nsp14, and the 2'-O-methyltransferase (2'-O-MTase) complex nsp10-16 (Bouvet et al. 2010 [↗](#); Chen and Guo 2016 [↗](#); Ramanathan, Robb, and Chan 2016 [↗](#)). The final steps of capping are the methylation of the 5'guanosine to form 7N-methyl guanosine ( $m^7G$ ) by nsp10-14 forming Cap0 and the methylation of the following nucleotide, often adenosine, at the 2'-O position to form Cap1 (Wilamowski et al. 2021 [↗](#)). Both methyltransferases use the substrate S-adenosyl methionine (SAM) as a methyl donor and form a functional complex with the enzymatically inactive nsp10

(Decroly 2011; Gupta 2021; Minasov 2020). They are part of the viral replication and transcription complex (RTC) consisting of enzymes responsible for RNA replication (nsp12), processing enzymes (nsp14, nsp13, nsp16) and co-factors (nsp7, nsp8, nsp10) (Malone et al. 2022 [↗](#)).

Early on, nsp10-16 has been identified as a potential drug target for SARS-CoV (Decroly et al. 2008 [↗](#); Chen et al. 2009 [↗](#); Byszewska et al. 2014 [↗](#)) and has emerged as a therapeutic target against SARS-CoV-2 (Chang and Chen 2021 [↗](#); Liu et al. 2010 [↗](#); Dong, Zhang, and Shi 2008 [↗](#)). It belongs to the class I methyltransferases and contains a Rossmann-fold, typical for methyltransferases, to bind SAM as the methyl donor and  $m^7$ GpppN-RNA (Cap0-RNA) as the methyl acceptor (Wilamowski et al. 2021 [↗](#)). The large active site of nsp16 can be divided into two main parts, the SAM and the Cap0 site (**figure 1** [↗](#)) which previously have been termed the low-affinity binding site (LBS) and high-affinity binding site (HBS), respectively (Rosas-Lemus et al. 2020 [↗](#)). In the context of this study, we further divide the SAM site into an adenosyl and a methionyl cavity (**figure 1B** [↗](#)). Several co-crystal structures of nsp10-16 with active site binders are available with ligands always binding into the SAM site and occupying both cavities (Krafcikova et al. 2020 [↗](#); Rosas-Lemus et al. 2020 [↗](#); Bobileva et al. 2021 [↗](#); Klima et al. 2022 [↗](#)). Unlike other compounds, the previously reported nsp16 inhibitor SS148 extends further at position N7 of the heterocycle beyond the SAM site, here referred to as the extended N7 pocket (**figure 1B** [↗](#)).

The second substrate binding site, here termed the Cap0 site, accommodates the Cap0 molecule of the adjunct viral RNA. The Cap0 site can also bind a  $m^7$ GpppN molecule which includes only the first nucleotide, preferably adenosine ( $m^7$ GpppA, Cap-analog). The Cap0 site is bracketed by two gate loops, i.e. gate loop 1 (Met6818 – Ile6838) and gate loop 2 (Met6929 – Asn6941) which are displaced upon binding of Cap0-analog or Cap0-RNA (Viswanathan et al. 2020 [↗](#)). With SAM and Cap0 bound to the active site, the SAM methyl group and 2'-OH group of the Cap0 adenosine are in close proximity allowing for methyl transfer (Viswanathan et al. 2020 [↗](#)). Additional ribonucleotides following the Cap0-adenosine increase the reaction efficiency, presumably because they induce a slight shift of the acceptor base resulting in a better position for the transfer reaction (Minasov et al. 2021 [↗](#)).

Few nsp10-16 inhibitors are known to date. Sinefungin is an analog of S-adenosyl-L-homocysteine (SAH), the by-product of the methyl transfer reaction. In biochemical assays sinefungin has been shown to inhibit nsp10-16 (Otava et al. 2021 [↗](#); Devkota et al. 2021 [↗](#); Yazdi et al. 2021 [↗](#); Perveen et al. 2021 [↗](#)) but it has a low inhibitory effect in cell culture-based assays (Bobileva et al. 2021 [↗](#); Bergant et al. 2022 [↗](#)). Poor membrane permeability due to its zwitterionic nature has been suggested to cause this low activity (Ferreira de Freitas, Ivanochko, and Schapira 2019 [↗](#); Bobileva et al. 2021 [↗](#)). In another study, the methionyl moiety of SAM was systematically exchanged and several compounds, mainly aromatic groups at this position, were identified as more potent than sinefungin but lacked selectivity (Bobileva et al. 2021 [↗](#)). Recently, tubercidin has been suggested as a nsp16 inhibitor (Bergant et al. 2022 [↗](#)). It is known to exhibit broad antiviral, antitrypanosomal, and antifungal effects but it is cytotoxic and lacks selectivity (Bergstrom et al. 1984 [↗](#); De Clercq et al. 1986 [↗](#); Eyer et al. 2016 [↗](#)). Klima et al. 2022 [↗](#) identified the adenosine and tubercidin derivatives WZ16 and SS148, respectively (**figure 2** [↗](#)), as nsp16 inhibitors with an  $IC_{50}$  in the low micro molar range and crystal structures have been solved. These tubercidin-derived compounds are less cytotoxic and decrease the activity of nsp16 in biochemical assays (Klima et al. 2022 [↗](#); Schultz et al. 2022 [↗](#); Li et al. 2023 [↗](#)) and in the case of SS148 also the activity of nsp14 (Devkota et al. 2021 [↗](#)). Despite the knowledge of binders and inhibitors together with diverse *in silico* drug screens (Xu et al. 2020 [↗](#); Maurya et al. 2020 [↗](#); El Hassab et al. 2021 [↗](#); Vijayan et al. 2020 [↗](#)) and an NMR fragment screen (Berg et al. 2022 [↗](#)), no selective and effective nsp16 inhibitor has been identified to this point. Crystal structures of nsp10-16 from SARS-CoV, SARS-CoV-2, and MERS-CoV are available (Wilamowski et al. 2021 [↗](#); Chen et al. 2011 [↗](#), e.g. PDB entry 5YN6) and significantly contribute to the understanding of substrate binding and conformational changes upon Cap0 binding, catalysis and product release. The number of co-crystal structures with small molecule inhibitors is, however, limited and no X-ray compound



**Figure 1**

### Nsp10-16 overview and active site architecture

Structural representation of nsp10-16 with nsp10 as blue and nsp16 as beige surface (based on PDB entry: 7JIB). A) The active site is highlighted with a black frame and shows the substrates SAM and Cap0-analog as stick models with carbon in gray, oxygen in red, sulfur in yellow, nitrogen in blue, and phosphate in orange. B) Schematic representation of the active site with ligands as stick models as in A). The relative positions of the gate loops and the sub division of the Cap0 and SAM site is shown as described in the text. The extended cavity at the SAM site extends from N7 at the adenosyl cavity.

screening has been undertaken unlike for other SARS-CoV-2 drug targets such as nsp14, PL<sub>pro</sub>, M<sub>pro</sub>, RdRp, and S glycoprotein (Günther et al. 2021 [↗](#); Manandhar et al. 2021 [↗](#); Kuzikov et al. 2021 [↗](#); Khalifa et al. 2021 [↗](#); Imprachim, Yosaatmadja, and Newman 2023 [↗](#)).

Here, we explore the potential binding space of the nsp16 active site using a dedicated crystallographic screen of 234 compounds with derivatives of the natural substrate SAM as well as adenine analogues including previously described methyltransferase inhibitors. The selection of about half of these compounds was guided by molecular docking and the selection of a diverse subset by a fingerprint-based 2D similarity analysis of a large compound library targeting the SAM site. Further, we elucidate the structure of nsp10-16 in complex with tubercidin, which has been suggested as a potential nsp16 binder. We probed the binding of three additional tubercidin derivatives to nsp10-16 – namely toyocamycin, sangivamycin, and 5-iodotubercidin. Among the identified ligands binding to the SAM site, we discovered the adenosine analogue W08. Out of the *in silico* selected compounds for screening, W08 is structurally one of the most similar compounds compared to SAM, however, it binds unexpectedly in an ‘inverted manner’ compared to other analogs and extends into the extended N7 binding pocket. We explored the binding and antiviral effect of tubercidin and its derivatives which all bind to the nsp16 active site and whose N7 modifications extend into the extended N7 pocket as well, retaining the conserved binding mode despite their modifications. Additionally, we obtained structures of toyocamycin and sangivamycin together with Cap0-analog and Cap0-RNA. It has been unclear if the entire SAM site needs to be occupied for the second substrate to bind. Toyocamycin and sangivamycin are lacking the methionyl moiety of SAM. The toyocamycin/sangivamycin Cap0-analog/-RNA structures show that the Cap0-binding site can be occupied independently of the methionyl moiety. Our results provide important information required to understand the structural variety of compounds that are able to occupy the nsp16 active site which will enable the design of more specific antiviral methyltransferase inhibitors in the future.

## Results

### SAM is replaced by derivatives of SAM, adenosine, and tubercidin

Crystals of nsp10-16 grew within one to two weeks to 75-150 μm size under crystallization conditions reported by Rosas-Lemus et al. 2020 [↗](#) and crystal structures with ligands were refined against X-ray diffraction data at resolutions between 1.6 to 2.2 Å (table S 1). In total, 234 compounds were screened including two libraries from ASINEX, selected SAM fragments, tubercidin derivatives and known nsp16 inhibitors (figure S1) and identified 10 hits in the SAM site. Additionally, we identified several hits in the Cap0 site which will be described in a future manuscript (table S 2). All structures show the same overall fold as previously described (Viswanathan et al. 2020 [↗](#); Rosas-Lemus et al. 2020 [↗](#)). In the absence of other bound compounds, clear SAM electron density is observed in the active site although no SAM was added. The capturing of SAM during protein expression in *E.coli* has been observed previously (Lin et al. 2020 [↗](#)). The soluble fraction of nsp10-16 contains about 60 % bound SAM, as confirmed by isothermal titration calorimetry (ITC) (figure S 2A). However, the high occupancy in the crystal (modeled as 100 %) suggests that only the SAM-bound form crystallized. Attempts to remove SAM from soluble nsp10-16 prior to crystallization by extensive dialysis using charcoal led to 80 % ligand free-enzyme (compared to 40 % without dialysis) as shown by ITC (figure S 2B). However, crystallization trials with this SAM-reduced nsp10-16 did not yield crystals even when supplemented with SAM. The compounds identified as nsp16 binders in the crystal screen were soaked into crystals containing SAM and partially or fully replaced SAM. Partial replacement was observed for the ligands adenosine triphosphate (ATP), adenosine diphosphate (ADP), and W08 (ASINEX ID BDH 33959089). The combined occupancy of SAM and the respective ligand sums up to 100 %, suggesting that all active sites of the enzyme in the crystal are occupied. The other

**Adenosine derivatives**





$R = -OH$        $-S-CH_3$       

*Adenosine*      *5'-Methylthioadenosine (MTA)*      *W08 (BDH 33959089)*





*AMP*      *ADP*      *ATP*






*S-adenosyl-L-methionine (SAM)*      *S-adenosyl-L-homocysteine (SAH)*      *Sinefungin*      *WZ16 (Klima et al.)*

**Tubercidin derivatives**





*Tubercidin*      *Sangivamycin*      *Toyocamycin*




*5-Iodotubercidin*      *SS148 (Klima et al.)*

### Chemical structures of SAM-site binders.

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identified SAM site binders fully replace SAM. The binding of all tested compounds induces slight changes in the backbone root-mean-square deviation of atomic positions (RMSD) with the highest flexibility in the gate loops (figure S 3).

## SAM site binders occupy the adenosyl cavity and bind to an extended pocket

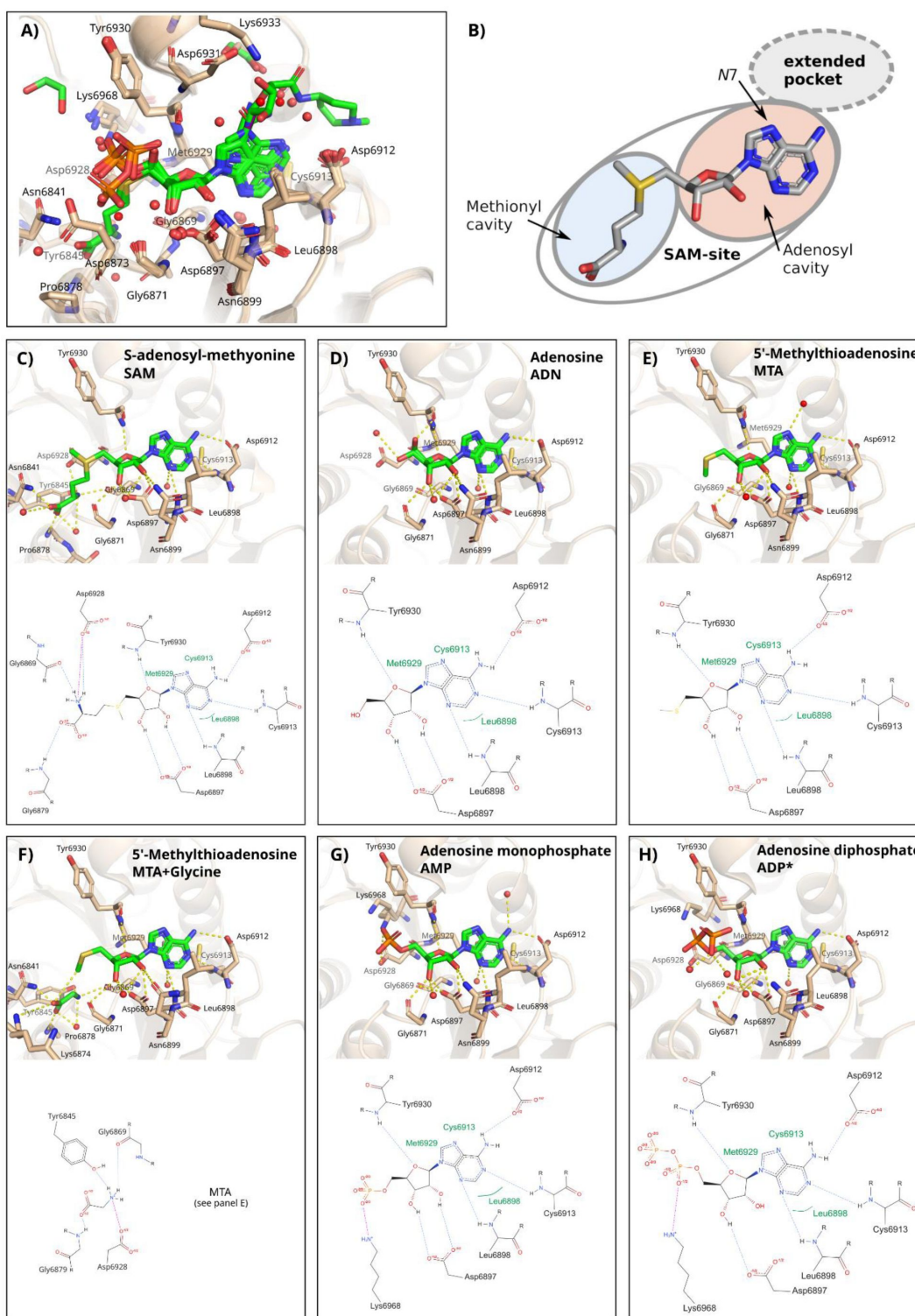
Adenosine, 5'-methylthioadenosine (MTA), and tubercidin are structurally similar (figure 2) and electron density is observed in the adenosine cavity while SAM is fully replaced (figure S 4). The active site interactions are identical to those of the adenosyl moiety of SAM where the following protein and ligand atoms are within hydrogen-bonding (H-bonding) distances: Asp6912 carboxyl oxygen and exocyclic nitrogen at C6; Cys6913 backbone nitrogen and N1; the carboxyl oxygens of Asp6897 and both ribose hydroxyl groups; Tyr6930 backbone nitrogen with cyclic oxygen of the ribose (figure 3). Additionally, the backbone carbonyl oxygen of Tyr6930 is within H-bonding distance to the 5'-OH of tubercidin, adenosine and MTA. These hydroxyl groups show alternative conformations modeled with almost equal occupancies either in the 'up' (in close proximity to Tyr6930, occupancy: 40 %) or in the 'down' position (occupancy: 60 %) with the oxygen at the same position as the sulfur atom in SAM (figure 3).

Furthermore, Met6929 might be involved in hydrophobic contacts with C2 of the purine of tubercidin and adenosine. In structures including SAM, this residue is in close proximity to the SAM sulfur atom. One to two water molecules are near to the exocyclic nitrogen of tubercidin in the 'down' conformation and might be involved in additional H-bonding. The binding mode of tubercidin agrees with the proposed *in silico* binding pose by Bergant et al. 2022.

To understand whether the binding of SAM can be imitated by two individual smaller fragments, we sequentially soaked MTA and glycine, resulting in ligand electron density for both compounds. In this structure, the orientation and interactions of MTA with the active site residues are identical to the structure of MTA alone (figure 3E, F). Glycine binds to the methionyl cavity which is occupied by an ethylene glycol in the structure with MTA alone. The potential interactions of the bound glycine with the protein are the same as the glycine corresponding part of the methionyl moiety of SAM. Hence the presence of glycine in the active site is most likely, showing that SAM binding can be imitated by the fragments MTA and glycine.

The tubercidin derivatives toyocamycin, sangivamycin and 5-iodotubercidin share the same orientation as tubercidin and the above-mentioned possible interactions with surrounding residues and waters are preserved (figure 3 J-M). None of the 5'-substituents of these derivatives have any surrounding interacting residues except for the amide group of sangivamycin that is in H-bonding distance of an ethylene glycol and a water molecule (figure 3K) or three water molecules in the case of the nitrile group of toyocamycin (figure 3L). The electron densities of the 5'-OH groups of 5-iodotubercidin and sangivamycin are interpreted as in the 'up' position but the 'down' position cannot be excluded (figure 3K, M). Vice versa, toyocamycin was modeled in 'down' position although the 'up' position cannot be excluded (figure 3L). The positions of tubercidin derivatives are very similar except for the aromatic heterocyclic ring of sangivamycin that is slightly shifted, resulting in a 0.4 Å offset of the 5'-C compared to tubercidin. The neighboring residue Asp6931 has also shifted away by 0.2 Å in this structure compared to tubercidin. In summary, a strong effect of 5'-substituents on active site residues could not be observed. This coincides with the observations for the binding and protein-ligand interaction of W08.

W08 is an adenosine analog and was on a low rank in the docking studies but is the third most similar compound to SAM in the library of docked compounds from ASINEX. In the crystal structure, electron density is present for W08 (occupancy: 57 %) and SAM (occupancy: 43 %). Surprisingly, the binding pose of W08 shows a 180° rotation compared to the SAM scaffold. As a

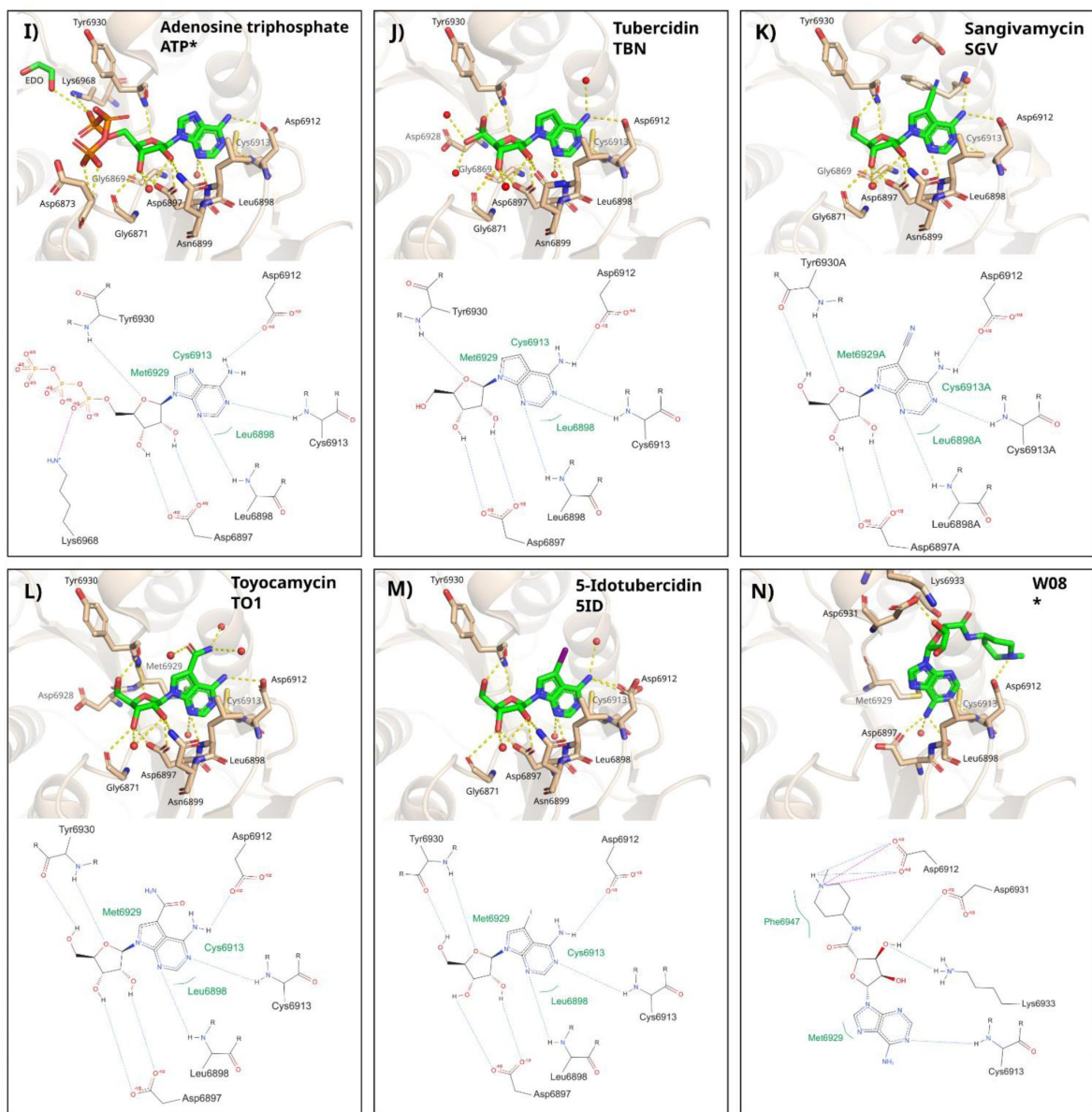


**Figure 3**

### Overview of interactions of ligands with nsp10-16.

A) Overlay of ligands (stick representation) bound to SAM binding site of nsp16 (cartoon representation). B) SAM site architecture adapted from [figure 1 B](#). C) - M) interaction of different ligands with nsp16. The top of each panel shows the ligand poses in the crystal including possible interactions as yellow dotted lines. On the bottom, possible interactions as 2D-plots only including relevant residues without the protein backbone are shown. Plots were created and edited with PoseView and PoseEdit (<https://proteins.plus>, Schöning-Stierand et al. 2022; Diedrich et al. 2023). Complex structures with partial occupancy by SAM are marked with an asterisk. For clarity, SAM is not shown even if present in these crystal structure.





**Figure 3** (continued)

consequence of the rotation, the ribose moiety points out into the extended N7 pocket towards the same position as the N7 substituents of the tubercidin derivatives (**figure 3N** [↗](#)). The contacts between the adenosine moiety of W08 and the active site residues differ from those observed with SAM (**figure 3A, N** [↗](#)). The exocyclic nitrogen of W08 is in close proximity to the carbonyl side chain atoms of Asp6897, whereas in SAM complexes the carboxyl moiety is in contact with the nitrogen of Asp6912. The cyclic N1 is, similar to SAM, in H-bonding distance to the backbone nitrogen atom of Cys6913. In the extended pocket, occupied by the piperidinamide ribose of W08, the 4'-OH group of the ribose is in H-bonding distance to the carbonyl oxygens of Asp6931 and Lys6933; the cyclic amine of the substituent is close to a carboxyl oxygen of Asp6912 and Phe6947. The latter might be involved in hydrophobic contacts with the piperidine ring (**figure 3N** [↗](#)).

The adenosine phosphates AMP, ADP and ATP were included in the screen to probe the Cap0 binding site as these are natural 'fragments' of Cap0 (**figure 2B** [↗](#)). Ligand electron density was, however, only observed in the SAM-binding site (figure S 4E-G). Their adenine moiety aligns with the equivalent substructure of SAM (**figure 3A** [↗](#)). The phosphates of AMP, ADP, and ATP are flexible and extend out of the SAM pocket and do not occupy the methionine cavity (**figure 3G-I** [↗](#)). In all cases, Lys6968 is in close proximity to the  $\alpha$ -phosphate oxygen and the  $\gamma$ -phosphate oxygens of the compounds that might also interact with the backbone nitrogen of Asp6873 (figure S 4E-G). Partial SAM density is still observed besides ADP (occupancy: 59 %) and ATP (occupancy: 58 %).

## In silico compound selection

For eight compounds with the highest ECFP4-based 2D similarity to SAM in the ASINEX library, electron density was not found in the SAM site after soaking experiments. Only W08 was identified as a SAM site binder and was selected based on its high 2D similarity to SAM is W08 which, however, binds in an unexpected orientation regarding the common molecular substructure of W08 and SAM.

Furthermore, no hits in the SAM site were found for the 93 compounds selected based on the molecular docking. For two compounds that include a theophylline substructure, partial electron density was found at the  $m^7$ G-position in the Cap0-site (results not shown). The docking with JAMDA (Flachsenberg et al. 2023 [↗](#)) led to compounds with higher scores than the one observed for SAM. However, the compounds are much larger than known binders and extend into the methionyl cavity or Cap0 binding site. Examples of poses are provided in figure S 5. We applied the crystal channel analysis tool LifeSoaks (Pletzer-Zelgert et al. 2023 [↗](#)) to see whether soaking of the proposed compounds might be hindered by narrow channels in the crystal. However, an analysis of PDB entry 8BSD shows that a broad channel with a 48.9 Å radius leads through the crystal and the binding site is freely accessible with an outer active site radius of 15.7 Å. The main differences between the selected compounds and known binders is the missing aliphatic character of the moiety attached to the ribosyl moiety of the compounds. Additionally, several compounds harbor other bicycle aromatic moieties instead of the adenine in SAM. Recent studies suggest that hydrophilic compounds are challenging regarding pose prediction with JAMDA (Flachsenberg et al. 2023 [↗](#)). This finding might explain why molecular docking did not lead to the identification of novel binders for nsp10-16.

## Replacement of SAM with tubercidin derivatives still allows binding of the Cap0- substrate

Compared to the native substrate SAM, tubercidin and its tested derivatives lack the methionyl moiety (**figure 2** [↗](#)). It is unclear if that part of the ligand in the SAM site affects the binding of the Cap0-RNA. To test if the second substrate, Cap0, can bind in the presence of selected SAM-site binders, toyocamycin and sangivamycin were soaked sequentially with Cap0-analog or Cap0-RNA ( $m^7$ GpppA UUAAA GGUUU AUACC UUCCC AGGUA) into nsp10-16 crystals. The resulting electron densities were compared to structures with SAM and Cap0-analog/Cap0-RNA. A structure

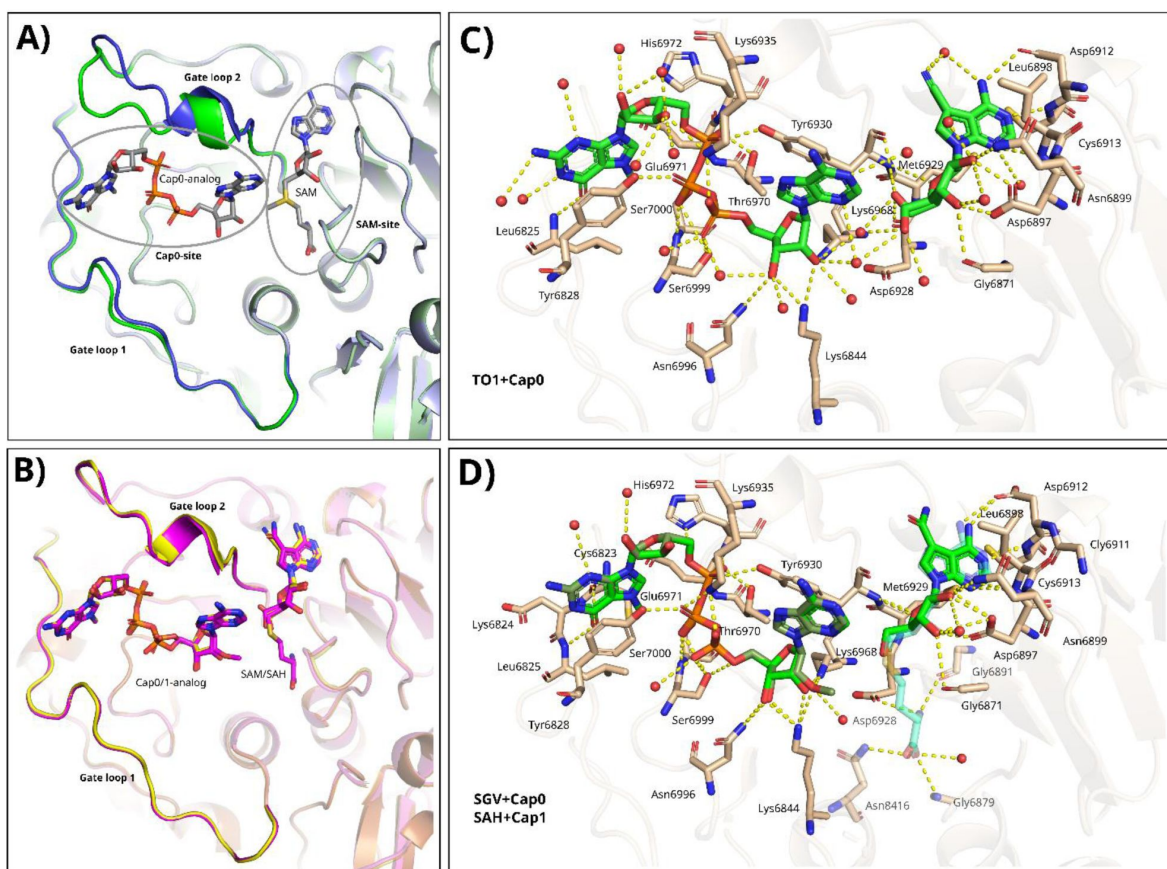
containing SAM and Cap0-analog has been published previously (PDB entry 7KOA, [Wilamowski et al. 2021](#)) but from crystals grown under different conditions. For better comparability with structures from our study, we obtained crystals containing the two substrates under the same conditions as ligand-soaked crystals. Here we found that ethylenediamine tetraacetic acid (EDTA) that had been used to slow down the methyl transfer reaction can also bind to the surface of nsp10-16 (figure S 6). More information on this finding can be found in the supplementary material. Nevertheless, the shift of the gate loops as seen in the substrate structures with SAM and Cap0-analog present (**Figure 4 A**) is observed when SAM is replaced by either of the tubercidin derivatives, toyocamycin or sangivamycin (**Figure 4B**). In all these structures the binding of the Cap0-analog and Cap0-RNA causes a shift of 1.5 Å in gate loop 1 for Tyr6828 (based on the backbone nitrogen position) and a shift of 2.4 Å of the Cα from Lys6935 and 2.1 Å of Cα from Pro6932 in gate loop 2 (**figure 4A, B**; S 2). There are no obvious differences in structures or interactions visible in the crystal structure from the tubercidin derivatives plus Cap0-analog/Cap0-RNA and the reference structure with SAM and Cap0-analog (**figure 4A, C, D**), suggesting that the binding of smaller compounds that do not occupy the methionine cavity also allows the native motions of the gate loops.

For toyocamycin together with Cap0-analog and Cap0-RNA, clear density is present (figure S 3M, N). Toyocamycin replaced SAM, although a remaining fraction of SAM cannot be excluded. Density in the allosteric binding site in the structure with Cap0-analog was interpreted as 7-methyl-guanosine-5'-triphosphate (MGP) as in other nsp10-16 crystal structures (PDB-entries 7JIB, 7JHE, 6WRZ, 6WVN). The crystal structure with toyocamycin and Cap0-RNA contains a Mg<sup>2+</sup> ion in the proximity of Cap0. In the case of sangivamycin soaked with the Cap0-analog and Cap0-RNA, the resulting densities are more ambiguous. The structure with the analog contains an overlay of Cap0-analog (occupancy: 46 %) and the product m7GpppAm (Cap1-analog) (occupancy: 54 %) and in the SAM site an overlay of sangivamycin (occupancy: 36 %) with SAH (occupancy: 64 %). In the structure with the Cap0-RNA, the RNA molecule was modeled with 100 % occupancy and in the SAM site sangivamycin (occupancy of 39 %) is present together with SAM (occupancy: 61 %).

## Tubercidin and its derivatives reduce viral titers

All identified binders of nsp16 (Adenosine, MTA, AMP, ADP, ATP, tubercidin, sangivamycin, toyocamycin, 5-iodotubercidin) were tested for their *in vitro* antiviral activity at 10 μM concentrations except for W08 since it was no longer available for purchase. SARS-CoV-2 infectious particles in the supernatant of treated infected Vero E6 cells were quantified via immunofocus assay while the effect of the compounds on cell viability was determined employing the cell counting kit-8 (CCK-8) method. Only tubercidin derivatives reduced viral titers by at least ten-fold, although at a clear expense of cell viability (figure S 7). Notably, sinefungin known as a pan-methyltransferase inhibitor and widely used as a control in nsp10-16 biochemical assays ([Otava et al. 2021](#); [Devkota et al. 2021](#); [Yazdi et al. 2021](#); [Perveen et al. 2021](#)) did not decrease virus replication. A plausible explanation for the lack of virucidal properties of sinefungin is its poor cell permeability ([Bobileva et al. 2021](#)).

To calculate the selectivity index (SI) of tubercidin substituents, we estimated their half-maximal effective concentration (EC<sub>50</sub>) and half-maximal cytotoxic concentration (CC<sub>50</sub>) in dose-response experiments. All tubercidin modifications reduced viral titers in a dose dependent manner (**figure 5**). However, due to its ability to reduce infectious particles by more than a hundred-fold exhibiting a SI greater than five, under our experimental conditions, we exclusively consider sangivamycin to have a selective effect on SARS-CoV-2 replication with an EC<sub>50</sub> of 0.01 μM (**table 1**). Although tubercidin reduces viral titers by more than hundred-fold, it presents limited therapeutic value due to an unfavorable toxicity profile demonstrated in cell-based assays by a SI of 3.75 (**figure 5**, **table 1**). Meanwhile, the reduction of viral titers detected at 10 μM and higher concentrations of toyocamycin and 5-iodotubercidin are largely the effect of cytotoxicity (**figure 5**, **Table 1**).



**Figure 4**

### Cap binding and gate loop motions.

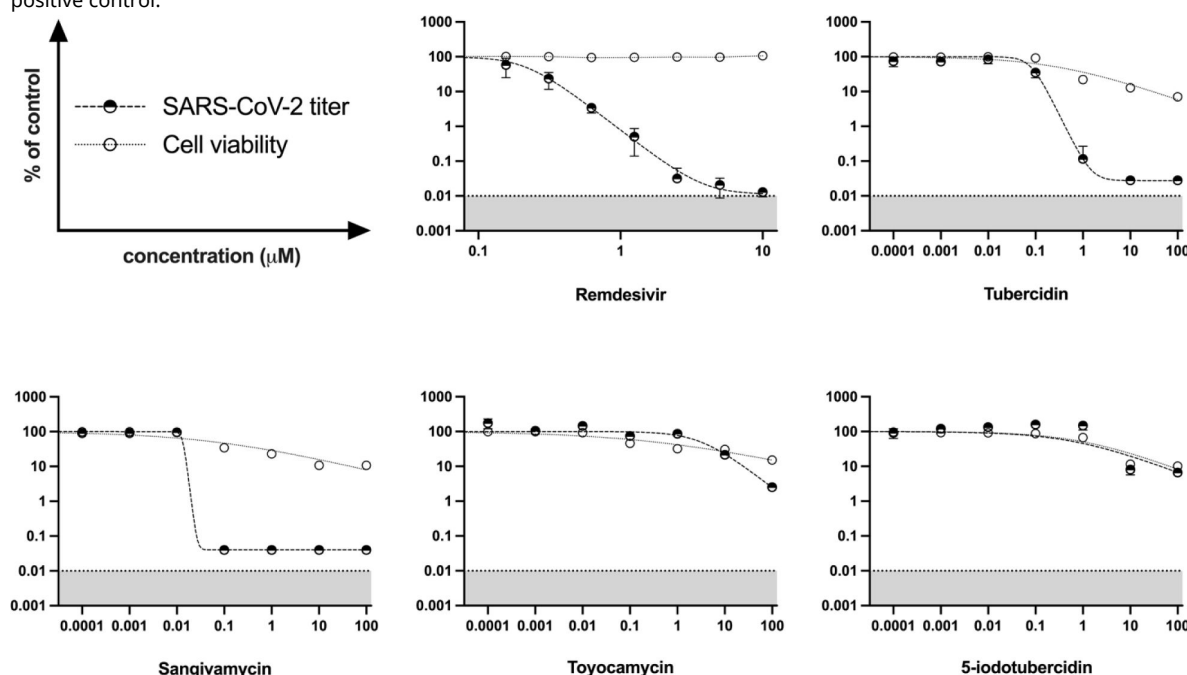
A) Gate loop conformations for nsp16 with bound SAM (blue ribbon) in overlay with the SAM and Cap0-analog bound structure (green ribbon). SAM and Cap0-site positions are indicated by circles. B) Overlay of nsp10-16 structures with bound toyocamycin and Cap0-analog (yellow ribbon and sticks) with the structure resulting from soaking sangivamycin and Cap0-analog (pink ribbon and sticks). In the latter structure, SAM and SAH are present aside from sangivamycin and a fraction of the Cap0- analog was converted into Cap1. The orientation is identical to A). C) Interactions of Cap0-analog and toyocamycin (TO1) with nsp16. D) Interactions of SAH/sangivamycin (SGV) and Cap0/Cap1-analog with nsp16.



**Figure 5**

**Effect of selected compounds on SARS-CoV-2 replication in Vero E6 cells.**

The viral titers (half-solid circles) and cell viability (empty circles) were determined by immunofocus assays and the CCK-8 method, respectively. Individual data points represent means  $\pm$  SD from three independent replicates in one experiment. Values below the methodological detection limit are shaded in gray. The dose-response curves were generated by fitting the data to the sigmoidal four parameter logistic function with  $1/Y^2$  weighting using GraphPad Prism version 9.5.1 for Mac OS X (GraphPad Software, San Diego, California USA, [www.graphpad.com](http://www.graphpad.com)). The maximum values were fixed to 100 %, and the minimum values were constrained to be equal to or greater than the methodological detection limit. Remdesivir was used as positive control.



**Table 1**

**Cytotoxicity, *in vitro* antiviral activity, and selectivity of selected compounds against SARS-CoV-2.**

CC<sub>50</sub>- half-maximal cytotoxic concentration; EC<sub>50</sub>- half-maximal effective concentration; SI- selectivity index. Cell viability and viral titers were determined by the CCK-8 method and immunofocus assays, respectively. Compounds that reduced infectious particles by at least a hundred-fold in combination with SI greater than five are considered antivirally active and are marked with an asterisk. Except for the SI, values were calculated from three independent replicates in one experiment by fitting the data to the sigmoidal four parameter logistic nonlinear regression function.

| Compound         | Cytotoxicity<br>CC <sub>50</sub> [μM] | Antiviral activity<br>EC <sub>50</sub> [μM] | Selectivity index<br>SI (CC <sub>50</sub> /EC <sub>50</sub> ) |
|------------------|---------------------------------------|---------------------------------------------|---------------------------------------------------------------|
| Remdesivir *     | >10                                   | 0.22                                        | >45.46                                                        |
| Tubercidin       | 0.3                                   | 0.08                                        | 3.75                                                          |
| Sangivamycin *   | 0.07                                  | 0.01                                        | 7                                                             |
| Toyocamycin      | 0.34                                  | -                                           | -                                                             |
| 5-iodotubercidin | 1.07                                  | -                                           | -                                                             |
| MTA              | >100                                  | -                                           | -                                                             |
| Sinefungin       | >100                                  | -                                           | -                                                             |



## Binding site comparison reveals structurally similar SAM pockets to nsp16 in human methyltransferases

Given the high cytotoxicity of the successfully soaked binders of nsp10-16, we set out to identify potential off-targets of the compounds. To this end, we applied SiteMine (Reim et al. 2024 [DOI](#)) to search for similar binding sites of SARS-CoV-2 nsp10-16 (figure S 8A) in structures of human methyltransferases. Figure S 8B shows the distribution of similarity scores obtained for searching in a methyltransferase-focused subset of the PDB (see Methods). The most significant similarities were found to SAM-complexed binding sites of the mitochondrial rRNA methyltransferase 2 (MRM2, UniProt accession: Q9UI43, PDB entry 2NYU; figure S 8C) and Cap-specific mRNA (nucleoside-2'-*O*)-methyltransferase 1 (HMTR2, UniProt Accession: Q8N1G2, PDB entry 4N49; figure S 8D). MRM2 fulfills crucial functions in the mitochondrial machinery and mutations in the enzyme cause MELAS-like clinical syndrome (Garone et al. 2017 [DOI](#)). The function of HMTR2 is poorly understood (Werner et al. 2011 [DOI](#)). Besides these two hits, we found 30 further enzymes with similar predicted or ligand-based sites. Several are critical in the human organism, e.g., tRNA methyltransferases and ribosomal rRNA processing enzymes. We visually inspected all binding site matches with a score of at least 31 (139 matches that can be clustered into 32 distinct proteins). Details regarding this analysis can be found in the supplementary material. We further investigated the similarities and differences between the SAM binding site of nsp10-16 and the best-scored hits (figure S 8). Intriguingly, we found highly conserved residues involved in hydrogen bond interactions with SAM (figure S 8). However, the surface and physicochemical properties in the direction of the *N*7 extension vary considerably (figure S 8). This finding might provide new starting points for the design of SAM-based selective inhibitors.

## Ligands bind to the nsp16 active site with low affinity

To assess ligand affinities for the nsp16 active site, a diverse set of methods was employed including nano differential scanning fluorimetry (nanoDSF), microscale thermophoresis (MST), ITC, ITC displacement assay, and spectral shift (NanoTemper Technologies). In brief, nanoDSF measurements with 10 nM protein and 1 mM ligands showed an increased stability of nsp10-16 with bound Cap0-analog (figure S 9C). SAM and SAH did not induce a thermal shift. Tubercidin and its derivatives seem to destabilize the protein complex (figure S 9A, B, D). Neither ITC (figure S 10) nor any other method tested showed measurable binding of the tested ligands within the investigated ligand concentrations applied, except for the substrates SAM and Cap0-analog. Spectral shift is not suitable for measuring nsp10-16 compound binding since no binding of the substrates SAM and Cap0-analog could be detected. In an attempt to measure the presumably weak binding of sangivamycin, an ITC displacement experiment was conducted where 2 mM sinefungin (as a “strong” binder) was titrated into 42.5  $\mu$ M protein that had been supplemented with 2 mM sangivamycin in advance. A control titration with sinefungin into protein was subtracted. The resulting  $K_D$  value of sangivamycin is  $> 1$  mM (figure S 11).

Taken together, despite clear density visible in difference electron density maps for ligands binding to the nsp16 active site, we could not determine exact apparent  $K_D$  values which are, however, estimated to be in the milli molar range.

In an effort to establish an MST displacement binding assay with SAM like molecules with an attached fluorophore, we obtained a crystal structure of nsp10-16 with the fluorescent RNA methyltransferase probe 5-FAM-triazolyl-adenosyl-Dab (FTAD) (Zimmermann et al. 2022 [DOI](#)) (figure S 12). More detailed information on binding assays and the FTAD crystal structure can be found in the supplementary material.

## Discussion

The lack of knowledge about binding fragments and small molecules to SARS-CoV-2 nsp10-16 has restricted the search and design of new and specific inhibitors. This study shows an approach similar to fragment screening with small drug or drug-like molecules to find new potential leads that can be further developed into selective inhibitors. Our screening approach using X-ray crystallography focused on SAM analogs but also included other smaller adenosine analogs like tubercidin and its derivatives. Unexpectedly, the compound W08 from an ASINEX library binds in a reverse orientation compared to all other identified binders and occupies an extended pocket of the SAM binding site instead of the methionyl cavity although it is one of the three most similar compounds to SAM in the library. This extended pocket might be a promising starting point to modify existing or develop future inhibitors to gain higher affinity and selectivity. Similar to W08, the tubercidin derivatives sangivamycin, toyocamycin and 5-iodotubercidin expand into this extended pocket, as do the already known inhibitors WZ16 and SS148 (Klima et al. 2022 [\[1\]](#)).

Despite clear compound electron densities in the nsp16 active site, binding constants could not be measured for tubercidin and tubercidin derivatives in any biochemical assay used in this study, except for sangivamycin which has a  $K_D$  of > 1 mM measured in an ITC displacement assay where sangivamycin was displaced by sinefungin. These compounds, however, showed different degrees of SARS-CoV-2 replicative attenuation in cell-based assays where all proved to be cytotoxic at a high nano molar range. Only the therapeutic efficacy zone of sangivamycin indicates a selective and potent activity against the virus. Tubercidin has been suggested as a nsp16 inhibitor. However, its antiviral effect has been characterized as a bispecific MTase inhibitor capable of reducing the enzymatic activity of the human cap methyltransferase 1 (HMTR2, UniProt: Q8N1G2) that can rescue SARS-CoV-2 mRNA methylation to a certain extent when the viral nsp16 catalytic function is impaired (Bergant et al. 2022 [\[2\]](#)). From the SiteMine binding pocket analysis HMTR2 has also been identified as one of the most similar enzymes to nsp16. Off-target effects on the viral site are very likely for 5-iodotubercidin and toyocamycin. 5-iodotubercidin has already been shown to target the multifunctional nsP1 that synthesizes Cap0 in the Chikungunya virus (Mudgal, Mahajan, and Tomar 2020 [\[3\]](#)), and it further inhibits SARS-CoV-2 RNA-dependent RNA polymerase (RdRp, nsp12). Toyocamycin can inhibit polymerases of other viruses (Bergstrom et al. 1984 [\[4\]](#); Wang and Yang 2020 [\[5\]](#)) but has not been tested yet against the SARS-CoV-2 RdRp. Another tubercidin derivative, 5-hydroxymethyltubercidin, can inhibit viral replication potentially by blocking the viral RdRp and with that the RNA synthesis of coronaviruses, including SARS-CoV-2 (Uemura et al. 2021). Targeting several viral proteins might prove beneficial for treatment, however, off-target effects in the patient should be avoided. Although restricted to publicly available structures, our analysis points towards more off-targets in human proteins of nsp10-16 inhibitors and highlights differences to be exploited in upcoming design studies to potentially reduce toxicity. On the other hand, the knowledge of similar sites might also help to develop novel ideas for selective inhibitors (e.g., PDB entry 6CKC - protein arginine N-methyltransferase 5 in complex with the selective and potent inhibitor LLY-283).

Although tubercidin derivatives may not target nsp16 specifically, these compounds can be used as starting points for further modifications to gain potency, specificity, and selectivity which has been demonstrated, e.g. by Klima et al. (2022) [\[1\]](#) with the tubercidin derivative SS148 that inhibits nsp16 activity ( $IC_{50}$  of 1.2  $\mu$ M with 1.7  $\mu$ M  $^3$ H-SAM and 0.8  $\mu$ M biotinylated-Cap0-RNA substrate concentrations). SS148 can be described as a merging product of toyocamycin and the homocysteine moiety of SAH. The native substrate SAM and the product SAH are built from adenosine or a methionine and homocysteine moiety, respectively. We identified adenosine as a binder in the active site of nsp16 in the crystal. Although we could not determine the binding affinity, its density is clearly visible in the crystal structure and can be used as a minimal building block for drug design.

The inhibitors sinefungin and its derivate WZ16 identified by [Klima et al. \(2022\)](#) [\[1\]](#) have detectable affinities and inhibitory activities in solution. They both occupy the methionyl cavity which might decrease compound affinity. [Bobileva et al. \(2021\)](#) [\[2\]](#) systematically replaced the methionyl moiety of SAM, resulting in several potent inhibitors that contain aromatic structures instead of the methionine. Especially *m*-benzoic acid derivatives show promising results despite lacking selectivity. These chemical scaffolds are valuable building blocks. In line with these observations, it is known that fragments and fragment-like compounds typically show weak binding and inhibition with  $K_D/K_I$  values in the micro molar to milli molar range ([Maveyraud and Mourey 2020](#) [\[3\]](#)). Although exceeding the aim of this study, a fragment merging approach might result in potent new inhibitors where the occupation of the methionyl cavity together with the extended pocket increases affinity and selectivity.

An intriguing approach is to target both the SAM and Cap0-site of nsp16. Several compounds selected from the docking study are larger than the known binders and were predicted to extend from the SAM to the Cap0 site. The crystallographic screening did not pick up any of these binders. W08, however, binds in an unexpected orientation without occupying the Cap0 site or methionyl cavity. In all reported structures with ligands bound to the SAM site, the Cap0 binding site is occupied by one or two molecules of ethylene glycol used as cryogenic protectant for the crystals. It cannot be excluded that the short exposure of the cryoprotectant after the long compound soaking displaced potentially bound compounds although it is unlikely. Efforts have been made previously to find bi-substrate binders for nsp16. It does not seem trivial to bridge the gap between the SAM and Cap0 sites with linkers as is possible for nsp14 ([Ahmed-Belkacem et al. 2020](#) [\[4\]](#); [Jung et al. 2022](#) [\[5\]](#); [Kottur et al. 2023](#) [\[6\]](#)). Since the extended pocket is adjacent to gate loop 2, compounds with large *N7* extensions that populate this pocket and block the movement of gate loop 2 might be an alternative to bi-substrate inhibitors. This might prevent binding of the Cap0 substrate and, thereby, inhibit the enzyme function.

The mechanism of the Cap0 substrate binding and the simultaneous gate loop motions are not fully understood. For example, it has been unclear whether the occupation of the entire SAM site is needed to allow the gate loop movements necessary for binding Cap0-RNA which is observed in crystal structures including SAM and Cap0-RNA or the Cap0-analog ([Minasov et al. 2021](#) [\[7\]](#); [Viswanathan et al. 2020](#) [\[8\]](#); [Wilamowski et al. 2021](#) [\[9\]](#)). Both loops shift by several Å upon binding of a Cap0 molecule. These dynamic changes are also observed in the present study when protein crystals were soaked with Cap0-analog or Cap0-RNA and SAM was replaced by sangivamycin or toyocamycin, which lack a methionyl moiety. The gate loop movements are very similar to what is observed in the SAM and Cap0-analog or Cap0-RNA, suggesting that ligand occupation of the methionyl cavity may not be a key factor for the gate loop motions.

Nsp10-16 contains multiple ligand binding sites but their dynamics, functions and potential for drug design are poorly understood. The phosphates of AMP, ADP, and ATP extend out of the SAM site near the position of the methionyl cavity. Due to the high flexibility of the phosphates, potential ligand-enzyme interactions are not clear. Nevertheless, this site could be explored for inhibitor design. Furthermore, as described recently, an additional cryptic pocket is located close to the extended *N7* binding pocket. It could be exploited as a druggable site, e.g. via fragment merging, since covalent and non-covalent binders prevent occupation by SAM and inhibit enzyme activity ([Inniss et al. 2023](#) [\[10\]](#)). We observed an EDTA molecule bound to the protein surface close to that position (figure S 6) which is also close to the *N7*-extension of W08. Another potential allosteric site with bound adenosine ([Viswanathan et al. 2020](#) [\[8\]](#)) or <sup>m7</sup>GTP (PDB entries 7JIB, 7JHE, 6WVN, 6WRZ) previously found in crystallographic studies has been suggested for drug targeting. This site is occupied by a buffer molecule (2-(*N*-morpholino)ethanesulfonic acid (MES)) in almost all structures reported here. These observations show that the potential ligand binding sites of nsp16 are still not fully understood and future studies will show their potential for drug discovery.

A significant complication concerning inhibitor discovery for nsp10-16 is that it is a multisubstrate enzyme. The binding-inhibition correlation is more complex than for single substrate enzymes and highly depends on the underlying kinetic mechanism (Sprenger et al. 2016 [DOI](#); Yung-Chi and Prusoff 1973 [DOI](#)). Furthermore, the SAM binding site is more than 50 % occupied with SAM when purified from *E. coli* and the protein becomes unstable upon removal of bound SAM, which challenges binding studies. Our results show that adenosine derivatives such as the adenosine phosphates can partially replace SAM and that the extended pocket can be occupied by several *N7* modified tubercidin derivatives. Such compounds should be explored further to gain higher affinity and selectivity of SAM-derived nsp16 inhibitors. Crystallographic screening is a very powerful tool to find new inhibitors and pharmacophore features especially for challenging targets such as nsp16. Further understanding the dynamics, regulation and kinetic mechanism of the nsp10-16 methyltransferase complex will provide crucial knowledge about the RNA processing of coronaviruses. Exploring the ligand binding sites presented and investigated herein will support drug development targeting SARS-CoV-2 and related viruses.

## Materials and Methods

### Protein expression, purification, and crystallization

Sequences of nsp10 and nsp16 (from polyprotein rep, 1a-1b uniprot ID: P0DTD1) were codon optimized for expression in *Escherichia coli* (*E. coli*) and synthesized and cloned into the pET-DUET-1 vector with a C-terminal Tobacco Etch Virus (TEV) protease cleavage site (ENLYFQG) and hexahistidine tag (His<sub>6</sub>) (GenScript). The expression construct was transfected into *E. coli* BL21 (DE3) pLyS (Promega, Madison, WI) and grown in Luria-Bertani broth (LB) supplemented with 50 µg/mL ampicillin and 130 µg/mL kanamycin overnight at 37 °C and 220 rpm. For inoculation of 1 L LB supplemented with ampicillin, 3 mL of the overnight culture were used. At OD<sub>600</sub> = 1.8 - 2 expression was induced by addition of 0.4 mM isopropyl-β-D-thiogalactopyranosid (IPTG) and further incubated at 18 °C for 16 h at 220 rpm. Cells were harvested by centrifugation for 1 h at 5.000 rcf and frozen until purification or used directly for protein purification. Cell pellets were resuspended in lysis buffer (50 mM Tris HCl, 500 mM NaCl, 2 mM MgCl<sub>2</sub>, 25 mM imidazole, 1 mM Tris-(2-carboxyethyl)phosphine (TCEP), 5 % (v/v) glycerol, pH 8.3) supplemented with EDTA-free protease inhibitor mix, lysozyme, and DNase. Cell lysate was sonicated with 40 % amplitude for 5 s on, 25 s off cycle for 25 min at 4 °C and cleared by ultracentrifugation at 120 000 rcf at 4 °C for 45 min. The supernatant was collected and loaded onto a His-Trap FF column containing a Ni-NTA resin using a GE Healthcare ÄKTA Pure system with loading buffer (50 mM Tris HCl, 500 mM NaCl, 2 mM MgCl<sub>2</sub>, 25 mM, 1 mM TCEP, 5 % (v/v) glycerol, pH 8.3). The column was washed with loading buffer and protein eluted with elution buffer (50 mM Tris-HCl, 500 mM NaCl, 500 mM imidazole, pH 8.3). The pooled protein fractions were dialyzed overnight in dialysis buffer (50 mM Tris HCl, 500 mM NaCl, 2 mM MgCl<sub>2</sub>, 1 mM TCEP, 5 % (v/v) glycerol, pH 8.3) at 4 °C and His<sub>6</sub>-tag was removed by adding TEV protease (3 mg per pellet out of 3 L expression culture). The cleaved tag was separated from the protein by Ni-NTA affinity chromatography and the protein further purified by size exclusion chromatography (Superdex 200 16/600 column) in crystallization buffer (10 mM Tris HCs, 150 mM NaCl, 2 mM MgCl<sub>2</sub>, 1 mM TCEP, 5 % (v/v) glycerol, pH 8.3). The protein was concentrated to 6 - 7 mg/mL (122 - 143 µM) and used immediately for crystallization or flash frozen in liquid nitrogen and stored at -80 °C.

### Crystallization and compound soaking

Protein crystals were grown by the hanging drop vapor diffusion method. For that, 1 µL protein with a concentration of 6.3 mg/mL and 1 µL reservoir solution (400 mM NaF, 100 mM MES at pH 6.5) were mixed. Rice shaped crystals of 75 - 100 µm length grew for approximately 19 days before they were soaked in reservoir solution supplemented with respective compounds. 5 - 20 crystals per compound were flash frozen in liquid nitrogen with 33 % ethylene glycol as cryogenic protectant. Compounds were dissolved to 100 mM in DMSO prior soaking and directly added to the

drop in volumes that result in 10 mM ligand concentration in the crystal drop. For soaking with only one compound, the crystals were harvested 24 h after soaking. To obtain the structures of toyocamycin and sangivamycin in the presence of Cap-0-analog ( $^{m7}$ GpppA purchased from New England Bioscience) or Cap0-RNA (Sequence:  $^{m7}$ GpppA UUAAA GGUUU AUACC UUCCC AGGUA, synthesized by BOCSCI INC, purified by HPLC, purity > 90 %), the compounds were first soaked for 24 h with 10 mM of tubercidin derivate and subsequently with 10 and 0.3 mM of Cap0 and Cap0-RNA, respectively.

## Selection of compounds for crystallographic screening

The entire list of compounds is documented in the supplementary material (table S 2). The selected compounds include a small compound library of SAM analogs purchased from ASINEX (80 compounds), known methyltransferase inhibitors, theophylline, tubercidin and adenine or adenosine derivatives as well as selected compounds from a docking study (figure S 4)

## Molecular docking and selection of adenosine derivatives from ASINEX purine library

The purchasable ASINEX compounds from the SAM analog, all nucleoside mimetics, and purine nucleoside libraries (3627 compounds) were prepared by UNICON (Sommer et al. 2016 [\[1\]](#)) using the best-scored protonation and tautomeric state per compound. We analyzed the crossdocking performance for the known SAM-site binders SAH, SAM, sinefungin and MTA, and the non-binders NTD008 and AZADC with an in-house version of the molecular docking tool JAMDA (Flachsenberg et al. 2023 [\[2\]](#)). To this end, we used the PDB entries 6W4H with SAM, 6W75 with SAM, 6WKQ with sinefungin, 6WQ3 with SAH, 6WRZ with SAH, 6WVN with SAM, 6XKM with SAM, 7JPE with SAM, 6WKS with SAM, 6YZ1 with sinefungin, 7BQ7 with SAM, 7C2I with SAM, 7C2J with SAM, and 7JYY with SAM. All structures were aligned to PDB entry 6W4H and the bound SAM was used as a reference ligand to define the binding site in a 6.5 Å radius. For the PDB entry 6YZ1, we observed the best cross-docking and ranking performance when retaining all water molecules considered as relevant in the docking preparation step with JAMDA. The molecular docking was performed with JAMDA default settings and 1000 conformations per compound. All compounds of the ASINEX library were docked in this manner and analyzed with PyMOL (Schrödinger and DeLano 2020 [\[3\]](#)). The best-scored 50 compounds were selected for crystal soaking experiments. In addition, 43 compounds with lower ranks were also selected based on visual inspection and a score at least similar to the one obtained for SAM. All selected compounds are considerably larger than known binders and extend into the pocket usually occupied by Cap0. They were purchased and tested for binding using soaking experiments.

## Diverse compound selection and selection based on similarity to SAM

In addition to the docking-based selection, we also selected a diverse set of 50 compounds based on the extended connectivity fingerprint (ECFP) with a diameter of 4 (ECFP4) (Rogers and Hahn 2010 [\[4\]](#)) and a distance-based k-Medoids clustering. The compounds assigned as medoids were purchased and tested for binding in soaking experiments. Additionally, the eight most similar compounds to SAM in terms of the ECFP4-based Tanimoto coefficient from the ASINEX set were selected for soaking experiments. All 2D similarity analyses were performed with KNIME [REF: 10.1145/1656274.1656280].

## Data collection, structure solution, and refinement

Data sets were collected at the PETRA III storage ring at Deutsches Elektronen-Synchrotron (DESY) in Hamburg at the P11 beamline and EMBL Hamburg beamlines P13 and P14. Data were processed using XDS (Kabsch 2010 [\[5\]](#)). As high-resolution limit of the data set, a  $CC_{1/2}$  above 0.3 – 0.5 was selected. The in-house developed data base AMARCORD (the AMARCORD system will be described



in a future publication) was used for data tracking and organization as well as a framework to run further scripts for processing and structure refinement. Within AMARCORD, data were processed using DIMPLE (Wojdyr et al. 2013 [\[1\]](#)) and refined with phenix.refine (Afonine et al. 2012 [\[2\]](#)) using the structure model of PDB entry 7JIB. The pipeline used for this was in principle the same as used in the DESY SARS-CoV-2 screening campaign on MPro (Günther et al. 2021 [\[3\]](#)), just omitting the clustering and PanDDA-step. Hit identification was done by visually inspecting the 2 mFo – DFc and mFo – DFc difference density maps. For the final structures, the refinement was done using phenix.refine with a template structure based on PDB entry 7JIB and subsequent model building and refinement was done by alternating rounds with COOT (Emsley et al. 2010 [\[4\]](#)) and phenix.refine. Data collection and refinement statistics are provided in table S 1.

## Structural SAM site comparison

We generated a GeoMine (Diedrich et al. 2021 [\[5\]](#)) database of potential human methyltransferases using all entries of the PDB (accessed: January 24<sup>th</sup>, 2024) in a complex with SAM or SAH. Additionally, we used all PDB entries with the GO annotation 0032259 (biological process: methylation). These entries were restricted to protein from *Homo sapiens*. We applied SiteMine (Reim et al. 2024 [\[6\]](#)) to compare the SAM binding site of nsp10-16 (PDB entry 7R1U, ligand three-letter-code of the binding site-defining binder: 4IK) to DoGSite3 (Graef, Ehrt, and Rarey 2023 [\[7\]](#))-predicted and ligand-based binding sites in the generated GeoMine database. The matches were ranked using the formula  $\text{SimilarityScore} = \frac{\text{PharmacophoreScore}}{\text{ShapeScore}} \times \text{SP} - \text{Score}$ . Default settings were used but the score normalization based on the larger pocket in terms of protein heavy atoms was disabled. The top-scored matches were visually inspected.

## Isothermal titration calorimetry

Affinities of compounds (Adenine, adenosine, AMP, ADP, ATP, MTA, 5-idotubercidin, toyocamycin, tubercidin, sangivamycin, sinefungin) to nsp10-16 were measured by ITC using a MicroCal PEAQ-ITC instrument (Malvern Panalytical). All titrations were performed at 25 °C, reference power of 10 µcal/s, stirring speed of 750 rpm, and a protein concentration of 45 µM in the sample cell and compound concentration of 500 µM in the syringe. All solutions contained the same buffer (50 mM Tris-HCl, pH 7.5, 200 mM NaCl). For each titration, one initial injection of 0.4 µL (0.8 s injection time) was followed by 12 or 18 injections with 2 µL each (4 s injection duration) and 150 s spacing between injections. Titrations of compound into buffer were performed as reference runs and subtracted from main runs. All thermograms were baseline-corrected and integrated using NITPIC (Keller et al. 2012 [\[8\]](#)). Data was fitted with a 1:1 binding model using SEDPHAT (H. Zhao, Piszczek, and Schuck 2015 [\[9\]](#)) and final plots were generated with GUSSI (Brautigam et al. 2016 [\[10\]](#)).

For the ITC displacement assay, 2 mM Sinefungin was titrated into 42.5 µM nsp10-16 mixed in advance with 2 mM sangivamycin. The buffer included 1 % DMSO. The  $K_D$  value for binding of sangivamycin was calculated with *ITCcalc* (Hammerschmidt et al. 2024 [\[11\]](#)).

## In vitro antiviral activity and cytotoxicity assays

Adenine, adenosine, AMP, ADP, ATP, MTA, sinefungin, toyocamycin, 5-idotubercidin, tubercidin, sangivamycin, and remdesivir were dissolved to 50 or 25 mM stock concentrations in either 100 % DMSO or water and stored at –20 °C. Vero E6 cells (ATCC CRL-1586) were grown in DMEM (PAN P04-03550) supplemented with 5 % FBS (PAN P30-3306), 1 mM sodium pyruvate (PAN P04-43100) and 1x antibiotic-antimycotic mix (ThermoFisher 15240062) at 37 °C and 5 % CO<sub>2</sub>.

Cells were seeded at  $2 \times 10^4$  cells/well in 96-well plates. After 24 h, the cell culture media was changed and serial dilutions of the compounds were added. Cell viability under 42 h compound treatment was determined via the Cell Counting Kit-8 (CCK-8, Sigma-Aldrich #96992) following the manufacturer's instructions.

Cells seeded at  $3.5 \times 10^4$  cells/well in 96-well plates were pretreated 24 h later with serial dilutions of the compounds. After 1 h incubation with the compounds, SARS-CoV-2 (strain SARS-CoV-2/human/DEU/HH-1/2020) was subsequently added at a multiplicity of infection (MOI) of 0.01 and allowed absorption for 1 h. The viral inoculum was removed, cells were washed with PBS without  $Mg^{2+}/Ca^{2+}$ , and fresh media containing the compounds (final DMSO concentration 0.5 % (v/v)) was added to the cells. Cell culture supernatant was harvested 42 h post-infection and stored at  $-80^\circ\text{C}$ . Titers of infectious virus particles were measured via immunofocus assay. Briefly, cells seeded at  $3.5 \times 10^4$  cells/well in 96-well plates were inoculated with 50  $\mu\text{L}$  of serial ten-fold dilutions of cell culture supernatant from treated cells. The inoculum was removed after 1 h and replaced by 1.5 % methylcellulose-DMEM (5 % FBS) overlay. Following incubation for 24 h, cells were inactivated and fixed with 4.5 % formaldehyde. Infected cells were detected using an antibody against SARS-CoV-2 nucleoprotein (ThermoFischer, PA5-81794). Foci were counted using an AID ELISpot reader from Mabtech. All experiments handling SARS-CoV-2 were performed at the biosafety level 3 (BSL-3) facilities in the Bernhard-Nocht Institute for Tropical Medicine in Hamburg under institutional safety guidelines.

The  $CC_{50}$  and  $EC_{50}$  values of the compounds were estimated by fitting the data to the sigmoidal four parameter logistic function with  $1/Y^2$  weighting using GraphPad Prism version 9.5.1 for Mac OS X (GraphPad Software, San Diego, California USA, [www.graphpad.com](http://www.graphpad.com)). The maximum values were fixed to 100 %, and the minimum values were constrained to be equal to or greater than the methodological detection limit. Samples deemed to be technical failures or extreme outliers were excluded from the calculations. The SI is defined as the ratio of  $CC_{50}$  over  $EC_{50}$ .

## Microscale thermophoresis to investigate nsp10-16 ligand interactions

MST experiments were performed as described previously for FTAD displacement (Zimmermann et al. 2022) and Tris-NTA Red labeling (Schwickert et al. 2022). Briefly, His<sub>6</sub>-tagged nsp10-16 (expressed from HEK cells; commercially obtained from BPS Bioscience Catalog #100747) was labeled using the Monolith His-Tag Labeling Kit RED-Tris-NTA second generation (NanoTemper Technologies), according to the manufacturer's instructions. Labeled protein was diluted to 10 nM into MST buffer (50 mM HEPES, 150 mM NaCl, 10 mM  $MgCl_2$ , 0.05 % polysorbate-20, 0.1 % PEG-8000, pH 7.5), and the ligands were added in a final concentration of 1 mM from DMSO. Measurements were performed in triplets at 25  $^\circ\text{C}$  and at medium MST power in Monolith NT.115 Capillaries (NanoTemper Technologies). The raw data were analyzed using the MO.Affinity Analysis software (NanoTemper Technologies).

## NanoDSF ligand screening

Thermal shift assays were carried out in triplicate on the Prometheus NT.48 nanoDSF instrument (NanoTemper Technologies) using the manufacturer-designated capillaries. Sample solutions contained 5  $\mu\text{M}$  protein and 10 mM of the investigated ligand in buffer (50 mM HEPES, 150 mM NaCl, 10 mM  $MgCl_2$ , 0.05 % polysorbate-20, 0.1 % PEG-8000, pH 7.5) and 10 % (v/v) DMSO. In the capillaries, the sample solutions were heated from 20  $^\circ\text{C}$  to 80  $^\circ\text{C}$  at a heating rate of 1.5  $^\circ\text{C}/\text{min}$ , and fluorescence was recorded at 330 nm and 350 nm. The measured fluorescence ratio of the detected fluorescence was plotted as a function of temperature using GraphPad Prism 7.04. The melting temperature was calculated as the inflection point of the resulting sigmoidal curve.

## Spectral shift assay

Spectral shift assay was performed on a Monolith X instrument (NanoTemper Technologies) in parallel with MST measurements. Nsp10-16 purified from *E. coli* was labeled using the Monolith Lys-Tag Labeling Kit RED-NHS second generation (NanoTemper Technologies), according to the manufacturer's instructions. Labeled protein was diluted to 25 nM into buffer (50 mM Tris-HCl, 200 mM NaCl, pH 8.3), and the ligands were added in a final concentration of 150  $\mu\text{M}$  SAM and 250

$\mu$ M Cap0-analog. Measurements were performed at 25 °C and at medium MST power in Monolith Premium Capillaries (NanoTemper Technologies). The raw data were analyzed using the MO.Affinity Analysis software (NanoTemper Technologies).

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## List of PDB entries

8BZV (ADN), 8OTO (AMP), 8OT0 (MTA+Gly), 8OV1 (ADP), 8OV2 (SGV), 8OV3 (SID), 8OV4 (TO1), 8C5M (MTA), 8OTR (W08), 8OSX (ATP), 8BSD (TBN), 9EMV (SGV+Cap0), 8S8W (SGV+RNA), 8S8X (TO1+Cap0), 9EMJ (TO1+RNA), 9EML (Cap0+EDTA)

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## Editors

Reviewing Editor

**Yang Yang**

Iowa State University

Senior Editor

**Amy Andreotti**

Iowa State University, Ames, United States of America

## Reviewer #1 (Public Review):

Summary:

This manuscript describes the crystallographic screening of a number of small molecules derived from the natural substrates S-adenosyl methionine (SAM) and adenine, against the SARS-CoV-2 2'-O-methyltransferase NSP16 in complex with its partner NSP10. High-quality structures for several of these are presented together with efforts to evaluate their potential biophysical binding and antiviral activities. The structures are of high quality and the data are well presented but do not yet show potency in biophysical binding. They only offer limited insights into the design of inhibitors of NSP16/10.

### Strengths:

The main strengths of the study are the high quality of the structural data, and associated electron density maps making the structural data highly accurate and informative for future structure-based design. These results are clearly presented and communicated in the manuscript. Another strength is the authors' attempts to probe the binding of the identified fragments using biophysical assays. Although in general the outcome of these experiments shows negative data or very weak binding affinities the authors should be commended for attempting several techniques and showing the data clearly. This study is also useful as an example of the complexities associated with drug discovery on a bi-substrate target such as a methyltransferase, several of the observed binding poses were unexpected with compounds that are relatively similar to substrates binding in different parts of the active site or other unexpected orientations. This serves as an example of how experimental structural information is still of crucial importance to structure-based drug design. In general, the claims in the manuscript are well supported by the data.

### Weaknesses:

The main limitations of the study are that the new structures generated in the study are fairly limited in terms of chemical space being similar to either SAM or RNA-CAP analogues. It feels a little bit of a lost opportunity to expand this to more diverse ligands which may reveal potential inhibitors that are distinct from current methyltransferase inhibitors based on SAM analogues and truly allow a selective targeting of this important target.

Another limitation is the potentially misleading nature of the antiviral assays. It is not possible to say if these compounds display on-target activity in these assays or even if the inhibition of NSP16/10 would have any effect in these assays. Whilst the authors do mention these points I think this should be emphasized more strongly.

### Minor critical points:

The authors state that their crystals and protein preps have co-purified SAM occupying the active site of the crystals. Presumably, this complicates the interpretation of electron density maps as many of the ligands share overlap with the existing SAM density making traditional analysis of difference maps challenging. The authors did not utilize the PanDDA analysis for this step, perhaps this is related to the presence of SAM in the ground state datasets? Also, occupancies are reported in the manuscript in some cases to two significant figures, this seems to be an overestimation of the ability of refinement to determine occupancy based on density alone and the authors should clarify how these figures were reached.

The molecular docking approach to pre-selection of library compounds to soak did not appear to be successful. Could the authors make any observations about the compounds selected by docking or the docking approach used that may explain this?

<https://doi.org/10.7554/eLife.98310.1.sa2>

### Reviewer #2 (Public Review):

#### Summary:

The study by Kremling et al. describes a study of the nsp16-nsp10 methyl transferase from SARS CoV-2 protein which is aimed at identifying inhibitors by x-ray crystallography-based compound screening.

A set of 234 compounds were screened resulting in a set of adenosine-containing compounds or analogues thereof that bind in the SAM site of nsp16-nsp10. The compound selection was

mainly based on similarity to SAM and docking of commercially available libraries. The resulting structures are of good quality and clearly show the binding mode of the compounds. It is not surprising to find that these compounds bind in the SAM pocket since they are structurally very similar to portions of SAM. Nevertheless, the result is novel and may be inspirational for the future design of inhibitors. Following up on the crystallographic screen the identified compounds were tested for antiviral activity and binding to np16-nsp10. In addition, an analysis of similar binding sites was presented.

#### Strengths:

The crystallography is solid and the structures are of good quality. The compound binding constitutes a novel finding.

#### Weaknesses:

The major weakness is the mismatch between antiviral activity and binding to the target protein. Only one of the compounds could be demonstrated to bind to the nsp16-nsp10 protein. By performing a displacement experiment using ITC Sangivamycin is concluded to bind with a  $K_d > 1\text{mM}$ . However, the same compound displays antiviral activity with an  $EC_{50}$  of 0.01 microM. Even though the authors do not make specific claims that the antiviral effect is due to inhibition of nsp16-nsp10, it is implicit. If the data is included, it should state specifically that the effect is not likely due to nsp16-nsp10 inhibition.

The structure of the paper and the language needs quite a lot of work to bring it to the expected quality.

#### Technical point:

Refinement of crystallographic occupancies to single digit percentage is not normally supported by electron density.

<https://doi.org/10.7554/eLife.98310.1.sa1>

#### Author response:

##### **eLife assessment**

*This important study describes the crystallographic screening of a number of small molecules against a viral enzyme critical for the 5' capping of SARS-CoV-2 RNA and viral replication. While the high-quality crystal structures and complementary biophysical assays in this study provide solid evidence to support the major claims regarding how these small molecule compounds bind to the viral enzyme, the mismatch between the antiviral activity and binding to the viral enzyme of several small molecule compounds could have been more thoroughly investigated or discussed. This paper would be of interest to the fields of coronavirus biology, structural biology, and drug discovery.*

We do fully agree that the antiviral assay results could be brought better into context clarifying that the antiviral effects of tubercine and its derivatives are due to off-target effects.

##### **Reviewer #1 (Public Review):**

###### *Summary:*

*This manuscript describes the crystallographic screening of a number of small molecules derived from the natural substrates S-adenosyl methionine (SAM) and adenine, against the SARS-CoV-2 2'-O-methyltransferase NSP16 in complex with its partner NSP10. High-*

quality structures for several of these are presented together with efforts to evaluate their potential biophysical binding and antiviral activities. The structures are of high quality and the data are well presented but do not yet show potency in biophysical binding. They only offer limited insights into the design of inhibitors of NSP16/10.

#### *Strengths:*

*The main strengths of the study are the high quality of the structural data, and associated electron density maps making the structural data highly accurate and informative for future structure-based design. These results are clearly presented and communicated in the manuscript. Another strength is the authors' attempts to probe the binding of the identified fragments using biophysical assays. Although in general the outcome of these experiments shows negative data or very weak binding affinities the authors should be commended for attempting several techniques and showing the data clearly. This study is also useful as an example of the complexities associated with drug discovery on a bi-substrate target such as a methyltransferase, several of the observed binding poses were unexpected with compounds that are relatively similar to substrates binding in different parts of the active site or other unexpected orientations. This serves as an example of how experimental structural information is still of crucial importance to structure-based drug design. In general, the claims in the manuscript are well supported by the data.*

#### *Weaknesses:*

*The main limitations of the study are that the new structures generated in the study are fairly limited in terms of chemical space being similar to either SAM or RNA-CAP analogues. It feels a little bit of a lost opportunity to expand this to more diverse ligands which may reveal potential inhibitors that are distinct from current methyltransferase inhibitors based on SAM analogues and truly allow a selective targeting of this important target.*

It is true that it makes sense to screen for more diverse compounds to expand to a more diverse ligand set and we do hope our study motivates to do so. Given the limited number of crystal structures of nsp10-16 with potential drug molecules, the aim of this study was to upgrade the data base with new complex structures to have a pool of complex structures for future compound designs with increased selectivity. Furthermore, some of the hits are known inhibitors of similar enzymes and most prominent and potent methyltransferase inhibitors are structurally related to SAM, like sinefungin and tubercidine. We do think that knowing which SAM compounds or fragments of SAM are able to bind in the nsp10-16 active site is highly valuable for further specific and optimized inhibitor design.

*Another limitation is the potentially misleading nature of the antiviral assays. It is not possible to say if these compounds display on-target activity in these assays or even if the inhibition of NSP16/10 would have any effect in these assays. Whilst the authors do mention these points I think this should be emphasized more strongly.*

That is a very valid point and we do not believe that the antiviral activity is based on on-target effects. We do agree that the way it is currently presented can be considered misleading and we indeed clarify this point in the revised version.

#### *Minor critical points:*

*The authors state that their crystals and protein preps have co-purified SAM occupying the active site of the crystals. Presumably, this complicates the interpretation of electron density maps as many of the ligands share overlap with the existing SAM density making traditional analysis of difference maps challenging. The authors did not utilize the*

*PanDDA analysis for this step, perhaps this is related to the presence of SAM in the ground state datasets? Also, occupancies are reported in the manuscript in some cases to two significant figures, this seems to be an overestimation of the ability of refinement to determine occupancy based on density alone and the authors should clarify how these figures were reached.*

We have used PanDDA in parallel for hit finding. We however did not see any advantages for this target over the hit finding results from the visual inspection. This is probably as mentioned because of SAM being present is the “ground state” which complicates the PanDDA map calculations.

Regarding the occupancies, we fully agree with this comment and change it to reasonable digits and clarify how the figures were reached.

*The molecular docking approach to pre-selection of library compounds to soak did not appear to be successful. Could the authors make any observations about the compounds selected by docking or the docking approach used that may explain this?*

Yes, it is a good point to give possible explanations why the docking approach was not successful to facilitate similar approaches in future studies.

#### **Reviewer #2 (Public Review):**

##### *Summary:*

*The study by Kremling et al. describes a study of the nsp16-nsp10 methyl transferase from SARS CoV-2 protein which is aimed at identifying inhibitors by x-ray crystallography-based compound screening. A set of 234 compounds were screened resulting in a set of adenosine-containing compounds or analogues thereof that bind in the SAM site of nsp16-nsp10. The compound selection was mainly based on similarity to SAM and docking of commercially available libraries. The resulting structures are of good quality and clearly show the binding mode of the compounds. It is not surprising to find that these compounds bind in the SAM pocket since they are structurally very similar to portions of SAM. Nevertheless, the result is novel and may be inspirational for the future design of inhibitors. Following up on the crystallographic screen the identified compounds were tested for antiviral activity and binding to np16-nsp10. In addition, an analysis of similar binding sites was presented.*

##### *Strengths:*

*The crystallography is solid and the structures are of good quality. The compound binding constitutes a novel finding.*

##### *Weaknesses:*

*The major weakness is the mismatch between antiviral activity and binding to the target protein. Only one of the compounds could be demonstrated to bind to the nsp16-nsp10 protein. By performing a displacement experiment using ITC Sangivamycin is concluded to bind with a  $K_d > 1\text{mM}$ . However, the same compound displays antiviral activity with an  $EC_{50}$  of  $0.01\text{ }\mu\text{M}$ . Even though the authors do not make specific claims that the antiviral effect is due to inhibition of nsp16-nsp10, it is implicit. If the data is included, it should state specifically that the effect is not likely due to nsp16-nsp10 inhibition.*

We do believe that the antiviral data are valuable and should be published within this work. We also agree with the comment that it should be clearly stated that the antiviral effect is not



likely because of nsp10-16 inhibition and we will optimize that accordingly.

*The structure of the paper and the language needs quite a lot of work to bring it to the expected quality.*

We will go through the manuscript again and further improve the structure and language as much as possible

*Technical point:*

*Refinement of crystallographic occupancies to single digit percentage is not normally supported by electron density.*

We agree with that point and correct it in the revised version.

<https://doi.org/10.7554/eLife.98310.1.sa0>